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This portfolio details three technical projects I led at Vecmocon Technologies and beyond, spanning IoT sensor analytics, EV infrastructure hardware, and connectivity engineering. Each case study follows a Problem → Solution → Challenges → Outcome structure, with emphasis on the engineering decisions, trade-offs, and delivery leadership behind each.

3 Technical Projects	IMU·BLE·GNSS Core Technologies	₹40L/mo Product Line Value	6+ Pilot Units Deployed
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PORTFOLIO INDEX

- ▶ **Project 1:** IMU-Based Crash Detection & Driver Behaviour Analytics (Vecmocon IoT product line)
- ▶ **Project 2:** Modular Battery Swapping Station — Built From Scratch (AUTOSAR-compliant)
- ▶ **Project 3:** Hybrid Geolocation & Find-My-Device — Wi-Fi / Cellular fallback + BLE (Apple & Google ecosystems)

IMU-Based Crash Detection & Driver Behaviour Analytics

Turning raw 6-axis motion data into safety-critical insights for EV fleets

6-axis

IMU Sensor Fusion

Real-time

Event Detection

Fleet-scale

Deployment

THE PROBLEM

Vecmocon's IoT product line powers electric two- and three-wheelers across India. Fleet operators and OEMs had a critical blind spot: they could not reliably detect crashes, harsh driving events, or risky driver behaviour in real time. This created several problems:

- ▶ **Safety:** Crashes went undetected for minutes or hours, delaying emergency response.
- ▶ **Insurance & Liability:** No objective data to reconstruct incidents or assess fault.
- ▶ **Fleet Risk:** Operators couldn't identify high-risk drivers (harsh braking, aggressive acceleration, sharp cornering) before accidents happened.
- ▶ **Battery & Vehicle Wear:** Aggressive driving degraded battery health and components faster, but there was no way to quantify or coach against it.

THE SOLUTION — WHAT I BUILT

I worked on the IMU-based analytics layer of the IoT product, transforming raw inertial sensor data into actionable safety events.

- ▶ **Crash Detection:** Defined the signal thresholds and multi-axis acceleration/gyroscope signatures that distinguish a genuine collision from routine bumps, potholes, or kerb-mounting.
- ▶ **Harsh Braking & Acceleration:** Built the event classification logic for harsh braking, rapid acceleration, and sharp cornering using deceleration profiles and jerk (rate of change of acceleration).
- ▶ **Driver Behaviour Scoring:** Aggregated event frequency and severity into a per-driver behaviour profile, enabling fleet operators to rank and coach drivers.
- ▶ **PM Ownership:** Translated these into PRDs and user stories for the engineering team, defined acceptance criteria, and prioritised which events to ship first based on customer (incl. TATA AutoComp) impact.

CHALLENGES & HOW I TACKLED THEM

⚠ Challenge 1: Distinguishing Real Crashes from False Positives

Approach: Potholes, speed bumps, and kerb impacts produce acceleration spikes that look like crashes to a naive threshold. I analysed real-world IMU datasets to characterise the signal signature of true collisions vs. road artefacts — looking at the duration, multi-axis correlation, and post-event motion (a real crash is usually followed by a stop; a pothole isn't).

Result: Defined a multi-condition detection rule that drastically cut false positives while retaining sensitivity to genuine crashes — making the feature trustworthy enough for fleet operators to act on.

⚠ Challenge 2: Defining 'Harsh' Objectively Across Different Vehicle Types

Approach: A harsh brake on a loaded e-rickshaw is different from one on a light scooter. A single hardcoded threshold would mislabel events. I worked with the engineering team to parameterise thresholds per vehicle class and used percentile-based calibration from real fleet data rather than arbitrary fixed values.

Result: Behaviour events became meaningful and comparable within each vehicle class, making driver scoring credible and actionable for fleet managers.

⚠ Challenge 3: Balancing Detection Accuracy vs. Device Compute & Bandwidth

Approach: Running heavy signal processing on a constrained embedded device, and streaming raw IMU data to the cloud, was neither power- nor bandwidth-efficient. I prioritised on-device event detection (sending only detected events + a short data window) over raw-data streaming, working with engineering to balance edge compute limits against detection fidelity.

Result: Achieved real-time detection within device constraints while minimising data costs — critical for a fleet-scale rollout across thousands of vehicles.

OUTCOME & IMPACT

- ▶ Delivered a safety-critical feature that became part of Vecmocon's differentiated IoT offering for fleet customers.
- ▶ Enabled objective incident reconstruction and driver risk profiling — directly supporting insurance and fleet-safety use cases.
- ▶ Demonstrated my ability to translate raw sensor physics into product features that customers pay for.

Modular Battery Swapping Station — Built From Scratch

An AUTOSAR-compliant MVP delivering 6 pilot units for field testing

From scratch Ground-Up Build	AUTOSAR Compliant MVP	6 units Field-Test Pilots
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THE PROBLEM

EV adoption in India is bottlenecked by charging time. Battery swapping — exchanging a depleted battery for a charged one in minutes — solves this, but requires reliable, safe, modular swapping infrastructure. The client needed a battery swapping station built from the ground up that could:

- ▶ Safely authenticate, eject, and accept battery modules without manual intervention.
- ▶ Manage state-of-charge (SoC) tracking and display for each slot.
- ▶ Control physical actuators (door locks, ejection mechanisms) reliably and safely.
- ▶ Meet automotive-grade software standards (AUTOSAR) for safety and maintainability.

THE SOLUTION — WHAT I BUILT

I led New Product Development for the swapping station as an end-to-end MVP, owning both the product definition and core feature execution.

- ▶ **Battery Swap Sequencing:** Defined and oversaw the full swap state machine — authentication, slot selection, door unlock, battery ejection, insertion, and re-lock — ensuring each step had safe fail states.
- ▶ **Door-Lock / Actuator Control:** Specified the control logic for physical actuators and interlocks, ensuring a battery could never be ejected unsafely or a door left unlocked mid-cycle.
- ▶ **SoC / User Interface Display:** Designed how state-of-charge and slot status surfaced to the user, so a rider knows instantly which slot to take and that the battery is fully charged.
- ▶ **AUTOSAR Compliance:** Ensured the MVP architecture followed AUTOSAR standards for automotive-grade reliability and future scalability.

CHALLENGES & HOW I TACKLED THEM

⚠ Challenge 1: Building Physical Infrastructure From Zero — No Reference Design

Approach: Unlike a software feature, this was a physical product with mechanical, electrical, and software subsystems that all had to integrate. I broke the system into independent modules (sequencing, actuator control, SoC display, safety interlocks) so subsystems could be developed and tested in parallel, then defined the interfaces between them up front.

Result: Delivered a working MVP despite no off-the-shelf reference, and the modular approach let us field 6 pilot units for client testing on schedule.

⚠ Challenge 2: Safety-Critical Actuator Control — No Room for Failure

Approach: A door unlocking at the wrong moment or a battery ejecting mid-charge is a serious safety hazard. I defined explicit fail-safe states for every actuator and built interlock conditions so that unsafe state transitions were physically impossible, not just discouraged in software.

Result: The station's swap sequence met both customer requirements and system safety constraints — a non-negotiable for field deployment.

⚠ Challenge 3: Meeting AUTOSAR Standards on an MVP Timeline

Approach: AUTOSAR compliance adds rigour but also overhead, which is hard to reconcile with MVP speed. I prioritised which AUTOSAR-compliant components were essential for the pilot vs. which could follow, keeping the core safety and architecture compliant while deferring non-critical polish.

Result: Shipped an AUTOSAR-compliant MVP that satisfied the client's automotive-grade expectations without blowing the pilot timeline.

OUTCOME & IMPACT

- ▶ Delivered 6 pilot units of a fully functional, AUTOSAR-compliant battery swapping station for client field testing.
- ▶ Owned a genuine 0-to-1 hardware product spanning mechanical, electrical, and embedded software domains.
- ▶ Proved I can lead New Product Development on safety-critical physical infrastructure, not just software.

Hybrid Geolocation & Find-My-Device

Keeping vehicles locatable when GNSS fails — Wi-Fi / cellular fallback + BLE

GNSS-denied Coverage Solved	WiFi+Cellular Fallback Positioning	Apple+Google BLE Ecosystems
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THE PROBLEM

The IoT product relied on GNSS (GPS) for vehicle location. But GNSS fails exactly where vehicles often are: underground parking, basements, dense urban canyons, and remote areas with poor sky visibility. This created two distinct problems:

- ▶ **Problem A — Location Blackout:** When GNSS was unavailable, the platform lost the vehicle's position entirely. Fleet operators couldn't track or locate vehicles parked in basements or operating in GNSS-denied environments.
- ▶ **Problem B — Finding a Specific Vehicle Nearby:** Even with rough location, a user standing in a large parking lot or crowded area needed a way to pinpoint and identify their specific vehicle at close range.

THE SOLUTION — WHAT I BUILT

I worked on two complementary connectivity features to eliminate the location blind spot.

A) Hybrid Geolocation (GNSS Fallback)

- ▶ When GNSS was unavailable or unreliable, the system fell back to Wi-Fi-based positioning (using surrounding Wi-Fi access point signatures) and cellular-based positioning (using cell tower triangulation).
- ▶ This gave a usable position estimate in basements, remote areas, and other GNSS-denied environments where the vehicle would otherwise have gone dark.

B) Find-My-Device via BLE (Apple & Google Ecosystems)

- ▶ Leveraged Bluetooth Low Energy (BLE) to enable a 'find my device' style feature for close-range vehicle location — letting a user pinpoint their specific vehicle nearby.
- ▶ Integrated with both the Apple (Find My) and Google (Find My Device) ecosystems so the feature worked natively for users across both phone platforms.

CHALLENGES & HOW I TACKLED THEM

⚠ Challenge 1: GNSS Gaps Left Vehicles Completely Untrackable

Approach: GNSS is accurate outdoors but useless in basements, tunnels, and dense areas. Rather than accept the blackout, I championed a layered positioning strategy: GNSS as primary, with automatic fallback to Wi-Fi and cellular positioning when GNSS confidence dropped. I defined when and how the system should switch sources.

Result: Vehicles remained locatable even in GNSS-denied environments, closing a major coverage gap that previously left fleet operators blind.

⚠ Challenge 2: Trading Off Accuracy, Power, and Cost Across Positioning Methods

Approach: Each positioning method has different accuracy, power draw, and data cost. Wi-Fi and cellular are less accurate than GNSS but work where GNSS doesn't. I defined a fallback hierarchy that used the most accurate available source while respecting the device's power and bandwidth budget — not running all radios simultaneously.

Result: Delivered continuous location coverage without unacceptable battery drain or data costs — essential for an always-on IoT device.

⚠ Challenge 3: Integrating Two Different BLE Ecosystems (Apple + Google)

Approach: Apple's Find My and Google's Find My Device networks have different protocols, privacy models, and integration requirements. Supporting only one would exclude half of users. I scoped the integration to work across both ecosystems so the find-my-device feature was platform-agnostic for the end user.

Result: Users on both iPhone and Android could locate their vehicle at close range natively — maximising the feature's reach and usefulness.

OUTCOME & IMPACT

- ▶ Eliminated the GNSS location blackout problem, making vehicles trackable in basements, remote areas, and dense urban environments.
- ▶ Delivered a cross-ecosystem (Apple + Google) BLE find-my-device capability for close-range vehicle location.
- ▶ Showcased my ability to architect layered connectivity solutions balancing accuracy, power, cost, and cross-platform reach.

WHY THESE PROJECTS MATTER FOR A PM ROLE

- ▶ Technical depth most PMs lack: I work fluently across sensor physics (IMU), automotive-grade hardware (AUTOSAR), and connectivity stacks (GNSS, Wi-Fi, cellular, BLE) — letting me earn engineering trust and make credible trade-off decisions.
- ▶ Trade-off thinking: Every project required balancing competing constraints — accuracy vs. power, speed vs. compliance, sensitivity vs. false positives — which is the core of product judgement.
- ▶ End-to-end ownership: From defining PRDs and acceptance criteria to overseeing delivery and field pilots, I owned outcomes, not just tasks.
- ▶ Customer impact: These features shipped into a ₹40L/month product line serving real fleet customers including TATA AutoComp Systems.

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